

Problem Statement Document

Project Title

AI-Powered News Trust and Credibility Analysis Platform

Domain

Artificial Intelligence (AI)
Natural Language Processing (NLP)
Deep Learning

Problem Statement

The rapid growth of social media, news websites, blogs, and messaging platforms has made it difficult for users to determine whether a news article is trustworthy or misleading.

Existing fake news detection systems mainly focus on classifying news as either "Fake" or "Real". However, these systems do not provide detailed credibility analysis, source verification, bias detection, multilingual support, or explanations for their predictions.

As a result, users may still find it difficult to understand why a news article is considered reliable or unreliable.

Therefore, there is a need for an intelligent platform that can evaluate the trustworthiness and credibility of news articles while providing transparent explanations to users.

Existing Work

Most existing fake news detection systems use:

- Machine Learning
- Deep Learning
- NLP Techniques

Common Algorithms:

- Naive Bayes
- SVM
- Random Forest
- CNN
- LSTM

- BERT

These systems mainly classify news as:

- Fake News
 - Real News
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Limitations of Existing Systems

1. Only classify news as Fake or Real.
 2. Do not analyze source credibility.
 3. Do not detect political or emotional bias.
 4. Do not verify information using multiple trusted sources.
 5. Do not explain the reason behind predictions.
 6. Limited support for regional languages.
 7. Do not provide detailed trust assessment.
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Proposed Solution

The proposed system aims to develop an AI-Powered News Trust and Credibility Analysis Platform that performs comprehensive analysis of news articles.

The system will include the following features:

1. Fake News Detection

Analyze news content using NLP and Deep Learning techniques to identify whether the article is fake or real.

2. Source Credibility Analysis

Evaluate whether the news source is reliable and trustworthy.

3. Bias Detection

Detect political bias, emotional manipulation, and one-sided reporting.

4. Clickbait Detection

Identify misleading headlines designed to attract user attention.

5. Cross-Source Verification

Compare important claims with trusted news sources to verify authenticity.

6. Explainable AI

Provide explanations for the credibility score generated by the system.

7. Multilingual News Analysis

Support multiple languages such as English, Tamil, Hindi, and Malayalam.

8. News Propagation Analysis

Track how news spreads across different platforms and identify the possible origin of misinformation.

9. Real-Time Breaking News Monitoring

Continuously monitor news websites and generate alerts when suspicious news articles are detected.

Research Gap

Most existing systems focus only on fake news classification.

Very few systems combine:

- Fake News Detection
- Source Credibility Analysis
- Bias Detection
- Clickbait Detection
- Cross-Source Verification
- Explainable AI
- Multilingual News Analysis
- News Propagation Analysis
- Real-Time News Monitoring

within a single platform.

Expected Outcome

The proposed system will help users:

- Identify trustworthy news articles.
- Understand bias in reporting.
- Verify important claims.
- Detect misleading content.
- Analyze the spread of misinformation.
- Understand the reasons behind credibility scores.

Conclusion

The proposed AI-Powered News Trust and Credibility Analysis Platform extends traditional fake news detection by incorporating credibility analysis, bias detection, multilingual support, cross-source verification, explainable AI, news propagation analysis, and real-time monitoring. This makes the system more reliable, transparent, and suitable for addressing modern misinformation challenges.